

# Samuel Monteiro Junior

## FULL-STACK & DEVOPS ENGINEER

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### PROFESSIONAL PROFILE

Full-stack engineer and DevOps and edge specialist in one: I deliver the product and the platform it runs on. I go from the frontend to the database (Next.js, React, FastAPI, Node, PostgreSQL) and take care of the infrastructure underneath (Docker, Cloudflare Workers, Tailscale, WireGuard, Caddy, hardening). One person covering the whole path: from idea to production to keeping it all running.

### TECHNICAL SKILLS

**Frontend:** React, Next.js, TypeScript, Vite, Tailwind CSS, Zustand, TanStack Query, Zod, Shadcn UI, Three.js

**Development & Automation:** Python (FastAPI, Web Scraping, CLI Scripts), Node.js (REST APIs, WebSockets, BullMQ, Redis), Rust (Tokio, Axum, Hyper), n8n Workflows, Shell/Bash Scripting for operational task automation

**DevOps & Infrastructure:** Docker, Docker Compose, Linux/Unix SysAdmin (Debian, Arch Linux, macOS), Caddy Server (Reverse Proxy, Automatic SSL), Mesh VPN (Tailscale, WireGuard), SSH Hardening, Networking & Firewalls (IPTables), Git

**Edge & Serverless:** Cloudflare Workers, Durable Objects, KV Store, D1, DNS-over-HTTPS, edge security heuristics

**Databases & Cryptography:** PostgreSQL, TimescaleDB, SQL Server, Redis Pub/Sub, SQLite, LUKS Full Disk Encryption, mTLS

### PROFESSIONAL EXPERIENCE

#### Automation Engineer & Full Stack Developer • Freelancer

April 2022 to Present

- Architected and deployed microservices ecosystems using FastAPI and Node.js in Docker containers, reducing request processing times by 40%.
- Designed and implemented a private mesh network (Tailscale VPN) with mTLS authentication to connect local hosts to Debian VPS servers, securing sensitive data traffic.
- Developed Cortex-Nexus, an automation pipeline (n8n, Python, AI) for lead extraction and qualification, achieving 99.7% deliverability and processing 3,000+ leads per month.
- Structured Parana-Leads, with automated deployment in Docker containers using Caddy as a reverse proxy with automatic SSL, reducing deployment time for new environments by 70%.

#### Support Analyst / Database Administrator • CNI Tecnologia e Software

June 2024 to December 2024

- Administered Linux/Windows network security policies and servers, reducing security incidents by 25% through connection hardening and access control.
- Developed Bash and Python scripts to automate relational database backups and hardware performance telemetry, reducing manual monitoring effort by 15 hours per week.
- Optimized queries and maintenance routines in SQL Server and PostgreSQL databases, reducing management report load times by 50%.

- Administered operational databases (PostgreSQL/SQL Server), implementing automatic backup routines and reducing downtime to near zero.
- Developed and maintained internal tools (Node.js/PHP) focused on query and load time optimization, accelerating internal API response times by 35%.
- Led the restructuring of IT support processes and technical documentation, resulting in a 30% reduction in repetitive support tickets.

## PROJECTS

### TessTrade: Algorithmic Trading Platform

[Rust](#) / [React](#) / [Python](#) / [TimescaleDB](#)

Live platform at [tesstrade.com](#) combining social features, a quantitative development environment, and backtesting: multithreaded Rust simulation engine, sandboxed Python indicators connected to the engine via FFI/RPC, and a React workspace with Monaco Editor for authoring strategies in the browser.

### 6ID: License Ecosystem for Trading Platforms

[FastAPI](#) / [Next.js](#) / [C# WPF](#) / [PostgreSQL](#)

Distributed ecosystem for license management and user authentication on trading platforms: asynchronous FastAPI backend (SQLAlchemy, Asyncpg), Next.js administration console with validation mirrored between Pydantic and Zod, and a Windows WPF client on .NET 8, compiled with ReadyToRun into a single binary.

### Aegis: Reverse Proxy and Rate Limiter

[Rust](#) / [Tokio](#) / [Axum](#) / [Docker](#)

Reverse proxy in Rust with per-IP token-bucket rate limiting, timeouts at every stage (including slowloris protection), SIGHUP configuration reload without dropping connections, Prometheus metrics on a separate admin plane, and a FROM scratch Docker image running as a non-root user.

### Sentinel: Edge Bot Firewall

[Cloudflare Workers](#) / [Durable Objects](#) / [JavaScript](#)

Edge bot firewall running on Cloudflare Workers and Durable Objects: each request is scored by TLS, HTTP version, and browser header signatures, and suspicious agents are blocked before reaching the origin, with no added latency and within the free tier.

### Cortex-Vault: Knowledge Infrastructure

[Linux](#) / [Shell Script](#) / [Python](#) / [Tailscale](#)

High-performance data ingestion system, automatically synchronized via Tailscale VPN mesh between Arch Linux and macOS. Developed Shell and Python utilities for fast local search and indexing.

### BoxOptimizer API

[.NET 8](#) / [JWT](#) / [Docker](#)

High-performance API for three-dimensional logistics calculation (efficient box cubing) using the First Fit Decreasing (FFD) algorithm, packaged as a Docker image and ready for horizontal scaling.

## EDUCATION

### Associate's Degree in Systems Analysis and Development

2024, ongoing

PUCPR (Distance Learning)

### Bachelor of Laws (LL.B.)

2016 to 2021

UNISEPE

### CC50: Introduction to Computer Science

2023

Harvard University / Fundação Estudar